TARUN L. SANKAR

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EDUCATION

University of Texas at Dallas — Dallas, TX Expected Graduation: May 2027

Bachelor of Science, Computer Engineering — Senior

Collegium V Honors • Dean's List • National Merit Scholar • GPA: 3.7/4.0

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash

Frameworks / Libraries: Next.js, React, Node.js, FastAPI, NestJS, Dash, TensorFlow, Manim

Infrastructure & Tooling: Azure, Docker, Kubernetes, Terraform, Harness, Argo, OpenObserve, FFmpeg, REST APIs

AI Systems: LLM tool calling, agent orchestration, multi-system context integration, visual-output evaluation

WORK EXPERIENCE

NorthMark Compute & Cloud (NMC²) — Dallas, TX Jun 2026 – Present

Software Engineering Intern, Network Automation

- Built and deployed a production Python alerting service used by 10+ network engineers, consuming OpenObserve webhooks and metrics to notify Slack of failed ZTP events and Infrahub changes; delivered 1,000+ alerts and surfaced 200+ ZTP errors.
- Designed a command-driven test harness that snapshots current Infrahub state as a baseline, validates subsequent changes against it, and posts configuration diffs to Slack when tests fail.
- Developed a tool-calling diagnostic agent that queries Jira, Slack, Infrahub, and ZTP systems to investigate open-ended network failures and correlate likely root causes; deployed the service through Harness and Argo-based CI/CD.

University of Texas at Dallas — Dallas, TX Aug 2024 – Present

Undergraduate Researcher

- Developed a real-time GNSS/IoT pipeline that computes S4 scintillation indices from satellite measurements sampled at up to 20 Hz, using POSIX C parallelization on a Raspberry Pi without interrupting receiver data logging.
- Reduced daily network transfer from 2.4 GB of raw measurements to 4.5 MB of processed indices (>99.8%), syncing results to Azure Blob Storage and refreshing a Dash monitoring dashboard every minute.

MyIntent.io — Remote Aug 2024 – Nov 2024

Software Engineering Intern, Full Stack Development

- Led UI/UX design for a dynamic, minimalistic business-facing web application.
- Built and shipped a production RFP automation tool (Next.js, NestJS) that uses Gemini and GPT-5 to generate business documents from structured customer inputs.

PUBLICATIONS & PRESENTATIONS

First Author — “Real-Time Scintillation Monitoring using Low-Cost GNSS-Based Monitors.” 2025 Joint CEDAR/GEM Workshop, Des Moines, IA. Poster

PERSONAL PROJECTS

Cinder Control — TAMUHack 2026 (2nd Place & Best Solo Project) Jan 2026

- Engineered a full-stack wildfire decision-support platform (Next.js, Python/FastAPI), solo, in 36 hours, letting incident commanders query fire behavior by voice or text.
- Built a parallelized cellular automata simulation projecting wildfire spread up to 96 hours out, cross-referenced against real infrastructure and geospatial data to flag at-risk assets and generate evacuation routes.
- Integrated OpenAI for natural-language control and ElevenLabs for voice narration, with Mapbox, OpenStreetMap, and NOAA NWS APIs powering the geospatial and weather layers.

Orune (useorune.com) — Generative Educational Animation Aug 2026

- Built and launched a full-stack generative animation platform that converts natural-language lesson prompts into narrated mathematical videos, with versioned renders, configurable generation depth, and MP4 export.
- Developed an iterative visual-review workflow that samples rendered frames, lets users draw and comment directly on problem areas, re-renders only affected scenes, and checks animated layouts for collisions before delivery.
- Implemented authenticated project history, revision management, concurrent generation controls, usage-based credits, Stripe billing, and optional Speechify narration.